

Hadis Farrokhi

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EDUCATION

University of Waterloo

Expected Aug. 2028

Bachelor of Software Engineering

Waterloo, ON

- Awards/Scholarships: President's Scholarship of Distinction
- Courses: Programming Principles, Introduction to Data Abstraction and Implementation

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, PHP, Bash, MySQL, HTML/CSS

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Scikit-Learn, TensorFlow, Flask, Node.js, React, FluidSynth

Tools & Technologies: Git, Docker, AWS, Jupyter, Codecov, REST APIs, VS Code

EXPERIENCE

Machine Learning Researcher

Jun. 2024 – Present

BCCC Research Center, York University

Toronto, ON

- Built a phishing detection model with **78% accuracy** using **Pandas** and **NumPy** to classify DeFi transactions
- Processed **1,000+ Ethereum logs** to extract features, using **Matplotlib** for pattern visualization and **regex** for parsing
- Implemented an advanced parser with **Flask** to build REST APIs and Python requests to automate data extraction

Software Engineer

Jan. 2025 – Present

Electrium Mobility

Waterloo, ON

- Collaborating with 20+ engineers to develop and optimize embedded firmware in **C/C++** for the OneWheel system
- Analyzing and modifying the Spintend Ubox **VESC** firmware to enhance motor control and **hardware integration**

Full Stack Developer

Sep. 2022 – Jun. 2024

RahWeb

Remote

- Deployed **10+** full-stack websites using **React**, **Node.js**, and **MySQL**, delivering solutions with **100% client satisfaction**
- Increased website traffic by up to **30%** using **SEO** strategies, focusing on keyword optimization and performance tuning
- Provided technical support, identifying and resolving website issues to maintain **99% uptime** and optimal performance

Volunteer Coding Instructor

Sep. 2023 – Jun. 2024

Connex Program

Toronto, ON

- Delivered workshops on **OOP fundamentals** in **Python** and **Java** to 50+ students, tailoring content to varied skill levels
- Guided students in coding projects by assisting with **white-box** and **black-box** testing to debug and optimize their code

President

Sep. 2023 – Jun. 2024

Math & Computer Science Club

Toronto, ON

- Directed training sessions for **CIMC**, **CTMC**, and **CCC** competitions, guiding 45+ participants to achieve top results
- **Two-time award-winning** club president for outstanding performance in improving students' math and coding skills

PROJECTS

Musical Keyboard | *Python, C, Raspberry Pi, WiringPi, FluidSynth*

Oct. 2024 – Jan. 2024

- Integrated **FluidSynth** with a **Raspberry Pi** keyboard implemented in **C**, delivering sound synthesis with **46ms** latency
- Enhanced user interaction by implementing custom **debounce logic** for **GPIO** handling, reducing input errors by **67%**
- Improved system stability by reducing crashes by **33%** through **GPIO** and audio optimization, backed by extensive testing

Resume Categorizer System | *Python, Pandas, Scikit-Learn, NLTK, Gensim*

Dec. 2023 – Jan. 2024

- Developed a resume categorizer with **86%** accuracy using **Pandas**, **NLTK**, and **Gensim**, trained on processed text features
- Achieved **87%** precision and **86%** recall by balancing the dataset with **SMOTE** and training a **Random Forest** classifier
- Generated interactive visualizations with **Matplotlib** and **Seaborn** to uncover key trends in data distribution and metrics

Destination Climate | *Python, Flask, API Integration, HTML, CSS, JavaScript*

Dec. 2024 – Jan. 2024

- Developed a **Flask** app that takes user-input temperatures and returns matching cities using a 3rd party API integration
- Designed a **responsive frontend** using **HTML**, **CSS**, and **Bootstrap**, providing seamless accessibility across devices
- Implemented exception handling and rollback mechanisms for reliable API responses and maintain application stability

Inclusive Workplace Calendar | *Python, Flask, JavaScript, HTML, CSS*

Nov. 2024 – Dec. 2024

- Developed a calendar app at WiE Hackathon using **Flask** and **JavaScript**, featuring holiday tracking from various cultures
- Created front-end interface using **JavaScript** to let users modify and filter holidays displayed dynamically on the calendar

CERTIFICATIONS

Machine Learning with Python - IBM 

2024

Python for Data Science, AI & Development - IBM 

2024

Crash Course on Python - Google 

2024